

1. A crane has a cable used to lift and move various objects. The cable is 300ft long and weighs a total of 2000lbs .
 - a) If half of the cable is currently out and needs to be retracted how much work is required to accomplish this?
 - b) If the cable is fully extended and is then hooked up to a 3-ton crate, how much work is required to retract the cable?

2. A 100ft length of steel chain weighing $15\text{ lbs} / \text{ft}$ is dangling from a pulley.

a) How much work is required to wind the chain onto the pulley?

b) How much work is required to wind half of the chain onto the pulley?

3. An 800ft cable weighs 1600lbs and is used to lift 800lbs of coal up a mine shaft which is 500ft deep. Find the work done by lifting all the coal to the top of the shaft.

4. A chain lying on the ground is 10m long and its mass is 80kg. How much work is required to raise one end of the chain to a height of 6m?

5. A cylindrical tank of radius $3ft$ and height $10ft$ is half filled with water. How much work is required to pump the water out of a hole in the top of tank? (The density of water is $62.4lbs / ft^3$)

6. A swimming pool is built in the shape of a rectangular prism which is 10ft deep, 15ft wide, and 20ft long.
- a) If the pool is filled with water to 1ft below the top, how much work is required to pump all the water into a drain at the top edge of the pool? (The density of water is $62.4 \text{ lbs} / \text{ft}^3$)

- b) A one-horsepower motor can do 550ft-lb of work per second. What size motor is required to empty the pool in 1 hour?

7. An inverted cone shaped water reservoir is $20m$ in diameter across the top and $15m$ in height. If the reservoir is filled to a depth of $10m$, how much work is required to pump all the water to the top of the reservoir? (The density of water is $1000kg / m^3$.)

8. The vat in the picture below contains water to a depth of 2m. Find the work required to pump all of the water to the top of the vat. (The density of water is $1000\text{kg} / \text{m}^3$).

